

Reyeal: Implicit Gaze-based Interaction for Creating Personal Landscape Painting in Cinematic Virtual Reality

Yang-Sheng Chen
National Taiwan University
Taipei, Taiwan
yeti0193275@gmail.com

Miguel Then Ying Jie
National Taiwan University
Taipei, Taiwan
miguelthen00@gmail.com

Chiao-En Hsieh
National Taiwan University
Taipei, Taiwan
rockjoe8107@gmail.com

Jing-Yuan Huang
National Taiwan University
Taipei, Taiwan
r11944029@ntu.edu.tw

Ping-Hsuan Han
National Taipei University of
Technology
Taipei, Taiwan
pinghsuan.han@gmail.com

Yi-Ping Hung
National Taiwan University
Taipei, Taiwan
hung@csie.ntu.edu.tw

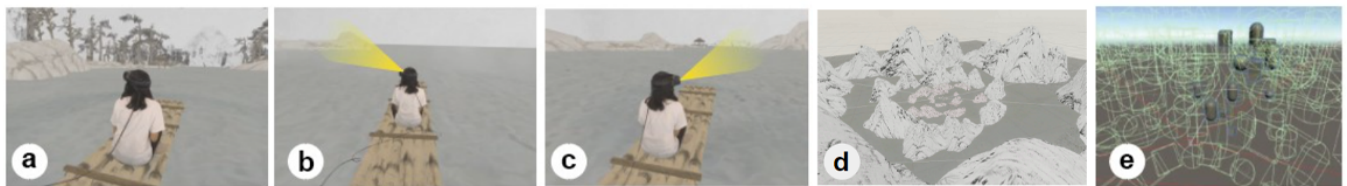


Figure 1: (a) We design an implicit gaze-based interaction in CVR experience, (b) the user can explore the virtual environment by gaze, (c) the objects in the scene reveal based on the user's gaze. (d) Overview of the virtual environment (e) To discuss the pattern of exploration and sense of control.

ABSTRACT

Cinematic virtual reality (CVR) brings viewers an immersive movie-watching experience. Different from storytelling of other media, CVR directors will use different interaction techniques and visual cues to guide the viewers through the immersive experience. In this paper, we present Reyeal, an implicit gaze-based interaction for creating personal landscape painting in cinematic virtual reality, in which the users will have their own experience of immersive landscape painting through their eyes. While most of the visual cues rely on visible objects, with Reyeal (Reveal by Eye), the CVR directors do not need to design interactions to guide the users looking at specific target objects. To demonstrate our technique, different elements of a Chinese landscape painting such as clouds and mountains will be drawn to the virtual environment by the users' gaze accordingly, which will result in a compelling experience for each user.

CCS CONCEPTS

• Human-centered computing → Virtual reality.

KEYWORDS

Cinematic Virtual Reality, Gaze-based interaction, Eye-tracking, Implicit interaction

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1 INTRODUCTION

With the advancement of head-mounted display (HMD) technology, many fields have begun developing VR applications. VR has also changed the viewing experience of movies. Cinematic Virtual Reality (CVR) allows viewers to watch films without the limitation of screen edges and controlling their own viewpoint. In addition to that, many CVR directors began to combine user interaction in the CVR experience. Hands-free interaction such as eye-based head gesture is a natural interactive technique that can enhance the exploration experience in CVR.

As previous works suggest, gaze-based interaction has been utilized in VR as an alternative for input from controllers. [Tanriverdi and Jacob 2000] firstly proposed gaze-based interaction in VR. They offered that gaze-based interactive technology can minimize physical effort and let users interact with distant objects in VR scenes easily. [Atienza et al. 2016] proposed designing the dynamic prompt panel and guidance butterfly by knowing where the VR user is looking and providing personalized guidance in the VR experience. CVR also utilized gaze-based interaction as a technique to enhance the

immersion of the users. "Rapture of the deep" [Probst et al. 2018] is a VR underwater experience that allows the VR environment to react to the user's gaze and attention to enhance the immersion. "Mise-Unseen" [Marwecki et al. 2019] is a software system that unnoticeably applies changes inside a VR user's field of view to provide dynamic staging in CVR. The work "Saw It or Triggered It" [Yang et al. 2021] explored the threshold of implicit and explicit interaction for the eye-tracking technique in CVR. GazeCues [Lankes and Ramirez Gomez 2022] proposed four types of gaze-based cues to enhance players' experience in VR exploration game.

Previous works mostly rely on visible objects and spaces to provide feedback for gaze interaction. In this work, we utilize the Implicit Gaze-based Interaction proposed in our previous work [Chen et al. 2023], which focused on designing the implicit interaction to reveal an invisible object that is supposed to be activated by the user's gazing attention. The goal is to let the users feel their gaze activates the object's appearance and enhances the immersion in the experience. This would also offer CVR directors with another interesting technique that can be used while designing CVR.

2 DESIGN AND IMPLEMENTATION

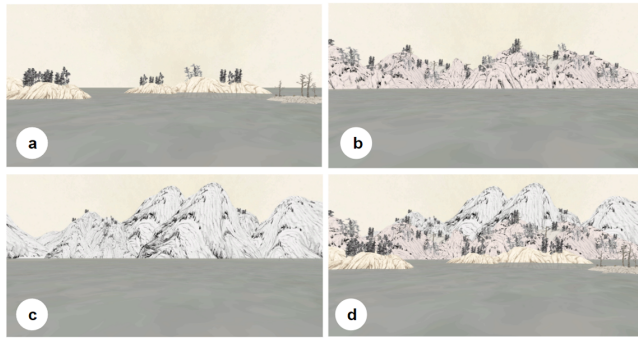


Figure 2: Virtual environment scene (a) near layer (b) middle layer (c) far layer (d) user's view

Our gaze-based interaction design consists of two elements: (1) Gaze-Based trigger method and (2) Visual feedback for triggered objects. For the Gaze-Based trigger method, apart from only using the traditional method of raycasting to trigger gaze interactions, we adjusted the triggering size of interactable objects based on objects to screen image ratio. For small interactable objects, we enlarge the triggering size for ease of trigger while maintaining the original triggering size for larger interactable objects to prevent the Midas touch problem. Visual feedback for triggered objects focuses on applying dissolve fade-in visual feedback from the bottom to top with a specific duration but in different ordering. In our design, the users' scene view (Figure 2d) consists of three layers, which are the near layer (Figure 2a), middle layer (Figure 2b) and far layer (Figure 2c). These layers are invisible initially which can be triggered respectively with users' gaze. We have arranged three types of orders: "Trigger-all", "near to far (NtoF)" and "far to near (FtoN)" for the experience. Trigger-all plays the animation for the interactable objects in one go within the trigger range. NtoF plays the animation of interactable objects from the nearest to the users, while FtoN plays the animation from the furthest object and then draws the small details one by one.

3 DEMO: IMMERSIVE LANDSCAPE PAINTING

To demonstrate the potential of implicit gaze-based interaction, we designed a three-minute CVR experience (Figure 1a), featuring a Chinese landscape painting. In this experience, users start in an initial scene that has only a blank sky with a Chinese rice paper texture and a boat in the middle of a lake. As the experience begins, the boat gradually moves across the lake to avoid motion sickness. Users are allowed to freely explore their surroundings (Figure 1b), interacting with the scene through their gaze. Various elements of the Chinese landscape painting, such as mountains, clouds, and trees, are triggered by our gaze-based interaction technique (Figure 1c). The users can choose their prefer triggering sequence for each experience. By the end of the experience, the virtual environment will be filled with different elements of the Chinese landscape painting (Figure 1d). Additionally, each user's personal experience can be recorded and archived. While the background such as mountains of the final virtual environment may remain the same, finer details such as trees, clouds, and rocks will differ based on users' interactions, making every immersive experience unique.

4 DISCUSSION AND FUTURE WORK

In our future work, we would investigate in the pattern of exploration of the users in virtual environment (Figure 1e) to design montage transitions and whether transitions triggered immediately after users' gaze would maintain the immersiveness during the experience. In this prototype, the interactable objects utilized non-photorealistic rendering, so the appearance for the interactive objects are unconstrained by reality. Therefore, we would also like to explore the possibility of applying this technique to physical-based rendering scenario, which will involves the design of transitions for interactable objects with photorealistic rendering.

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